# LUNAR LANDER

Implementation of a Genetic Algorithm in the Gymnasium Lunar Lander environment using the PyGAD library.

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# Goals and objectives

The main goal is to develop a genetic algorithm that, through an evolutionary process, learns to safely land on the lunar surface.

# ENVIRONMENT: LUNAR LANDER

## PHYSICAL CHALLENGES

- 8-dimensional state vector (position, velocity, angle).
- 4 discrete actions for engines steering

# WHY GENETIC ALGORITHM?

## GLOBAL OPTIMIZATION

Ability to avoid local minima through mutation mechanism

## GRADIENT FREE

GAs do not require a differentiable loss function, making them ideal for environments with sparse rewards

## LOGICAL SIMPLICITY

Nature-inspired approach allows for intuitive modeling of the learning process through "survival of the fittest"

# IMPLEMENTATION STRATEGY

## REPRESENTATION

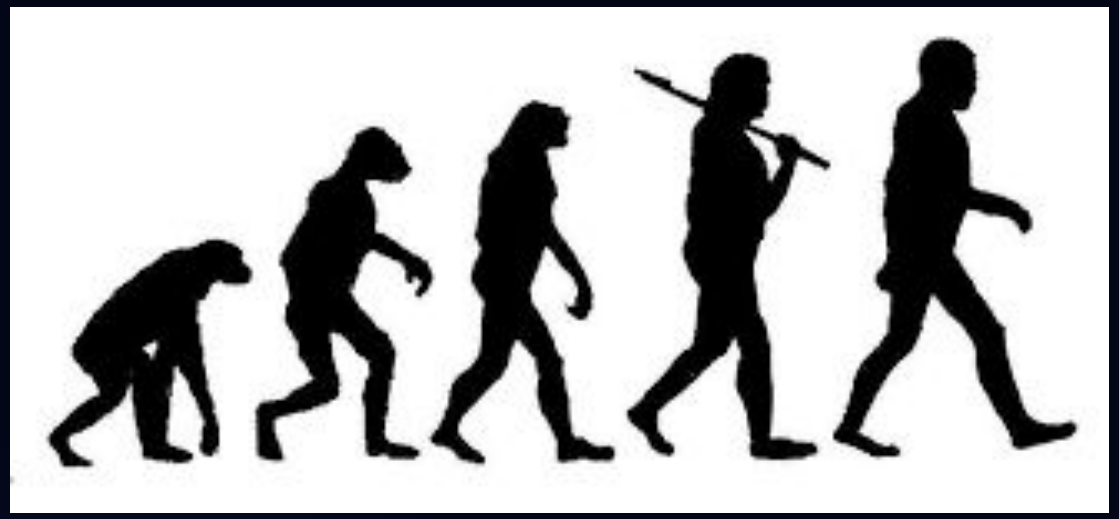
Each individual (chromosome) consists of a vector of floating-point numbers. These values act as direct control parameters (gains) for the lander's engines based on its current state (coordinates, velocity, angle)

## FITNESS FUNCTION

The performance of each controller is measured by the Total Reward from the Gymnasium Env.

- **Positive Fitness:** Awarded for a soft landing between the flags and stabilizing the aircraft.
- **Negative Fitness:** Penalties for crashing or flying out of bounds

# IMPLEMENTATION STRATEGY



## EVOLUTIONARY PROCESS

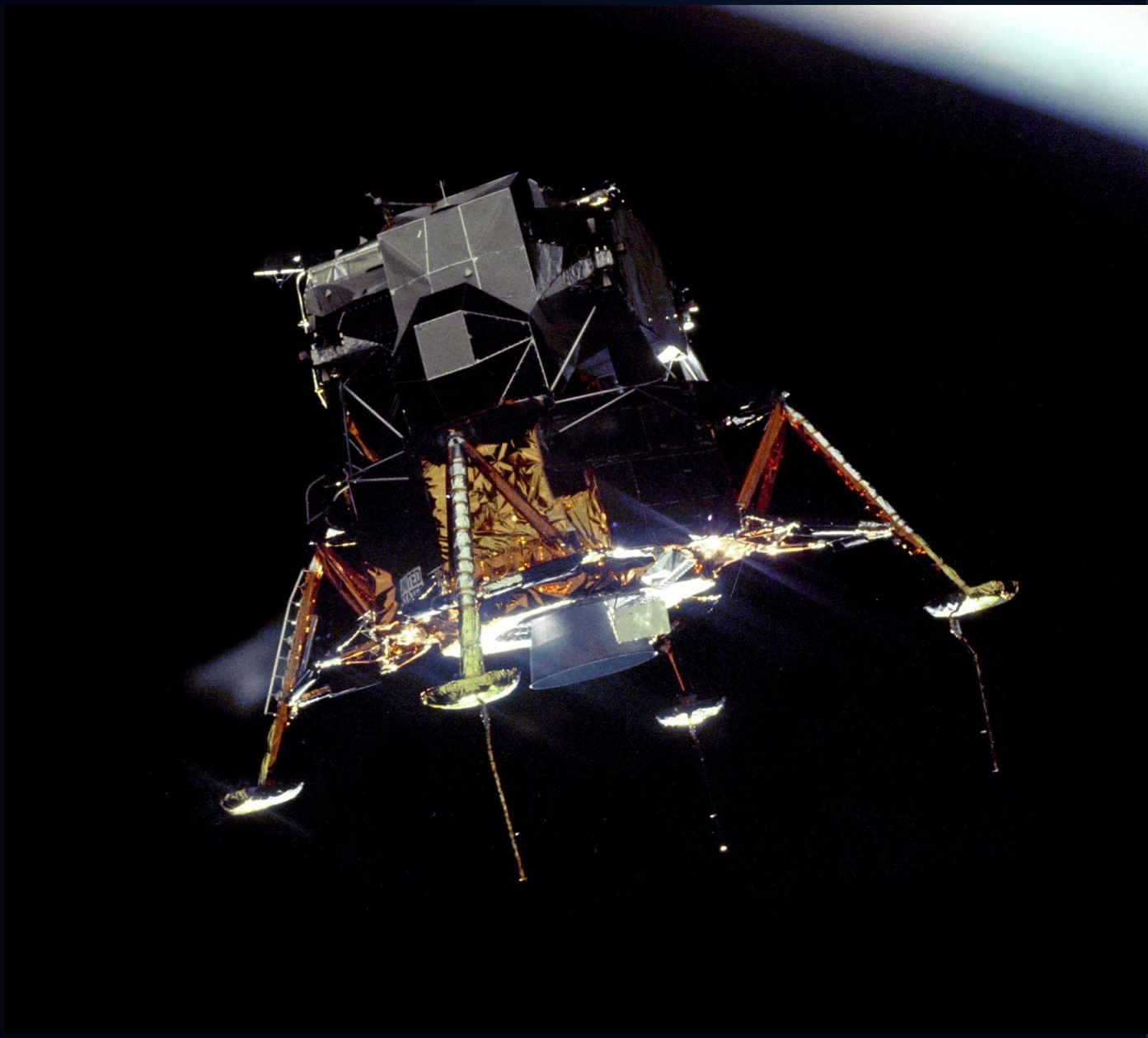
We would use PyGAD to manage the life cycle of the population:

- Selection: Identifying the top-performing "pilots" to act as parents
- Crossover: Combining parameters form two successful individuals to produce offspring.
- Mutation: Introducing random variations to the parameters to explore new landing maneuvers and prevent stagnation

## ENVIRONMENT INTEGRATION

The key challenge lies in integrating the PyGAD engine with the Gymnasium simulation loop. Each individual in the population will be evaluated based on the total reward accumulated during a single simulation run.

The final product will be a script demonstrating real-time landing using the evolved parameters.



# TECH STACK

## PYTHON 3.X

Base language providing flexibility and a rich ecosystem of scientific libraries.

## PYGAD

Optimization framework, offering ready-to-use genetic algorithm modules and support dor neural networks.

## MATPLOTLIB

Visualization tool for tracking evolutionary progress (fitness over generations).

## GITHUB

Version control system for hosting the project code and managing changes.

# EVOLUTION MECHANICS IN PYGAD

- **Fitness Function:** Defined as the sum of rewards from Gymnasium. The higher the score, the better the "individual.".
- **Selection:** Choosing the best population for reproduction.
- **Crossover:** The exchange of gens from fittest population
- **Mutation:** A random change genes, introducing new genetic information and preventing population stagnation.

# Thank you!

github.com/delux444/
genetic-lunar-landing